

1.Vectors.

i.Vectors type: Numeric, Character, Logical, Factor vector.

ii.Vectors function:

1.length().

```
x<-c(1,2,3,5)
```

```
length(x)
```

```
print(length(x)) #4
```

2.sum().

```
x<-c(1,2,3,5)
```

```
sum(x)
```

```
print(sum(x)) #11
```

3.mean().

```
x<-c(1,2,3,5)
```

```
mean(x)
```

```
print(mean(x)) #2.75
```

4.median().

```
x<-c(1,2,3,5)
```

```
median(x)
```

```
print(median(x)) #2.5
```

5.sort().

```
x<-c(5,4,3,2,1)
```

```
sort(x)
```

```
print(sort(x)) #1 2 3 4 5
```

6.reverse().

Opposite of sort().

iii.Vector Manipulation:

1.append().

```
x<-c(1,2,3)
```

```
y<-append(x,30)
```

```
print(y) #1 2 3 30
```

```
2.insert().  
x<-c(1,2,3)  
y<-append(x,30,2)==(variable,no.,index/position)  
print(y) #1 2 30 3  
3.remove().  
v1<-c(1,2,3,4,5,6)  
print(v1)  
print(v1[!v1%in%c(2,4)]) #[1] 1 2 3 4 5 6  
[1] 1 3 5 6
```

OR

```
v1<-c(1,2,3,4,5,6)  
print(v1)  
print(v1[v1%in%c(2,4)]) #[1] 1 2 3 4 5 6  
[1] 2 4  
4.unique().  
x<-c(1,2,3,4,5,6,4,3,5,6,1)  
print(unique(x)) #1 2 3 4 5 6  
5.duplicated().  
x<-c(1,2,3,4,5,6,4,3,5,6,1)  
print(duplicated(x)) # [1] FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE TRUE TRUE  
TRUE
```

iv.Vector Defination & Declaration.

```
x<-c(1,2,3)  
print(x) #1 2 3
```

v.Generating vector.

```
x<-c(1:5)  
print(x) #1 2 3 4 5
```

vi.Add or subtract vector.

```
x<-c(1,2,3)  
y<-c(4,5,6)  
print(x+y) #5 7 9
```

OR

```
x<-c(1,2,3)
```

```
y<-c(4,5,6)
```

```
print(x-y) #-3 -3 -3
```

vi.%in% operator.

```
x<-c(1,2,3)
```

```
y<-c(4,5,6)
```

```
print(x%in%y) #FALSE FALSE FALSE
```

OR

```
x<-c(4,5,6)
```

```
y<-c(4,5,6)
```

```
print(x%in%y) #TRUE TRUE TRUE
```

vii.Null variable.

```
x<-c(4,5,6)
```

```
is.null(x)
```

```
print(is.null(x)) #FALSE
```

viii.NULL

```
x<-c(4,5,6)
```

```
x<-NULL
```

```
print(x) #NULL
```

ix.Filtering & Subsetting.

```
x<-c(4,5,6)
```

```
x[x>4]
```

```
print(x[x>4]) #5 6
```

x.Excluding missing values.

```
x<-c(4,5,6,NA)
```

```
print(x[!is.na(x)]) #4 5 6
```

xi.Recycle vector.

```
b x<-c(1,2)
y<-c(4,5,6)
r<-x+y
print(r) #5 7 7
```

2.Matrices.**i.Creating a matrix.**

```
m<-matrix(c(1,2,3,4,5,6,7,8,9),nrow=3,ncol=3)
print(m) #
```

```
  [,1] [,2] [,3]
[1,]  1  4  7
[2,]  2  5  8
[3,]  3  6  9
```

ii.Add row and column. [row,column]

```
m<-matrix(c(1,2,3,4,5,6,7,8,9),nrow=3,ncol=3)
m<-rbind(m,c(10,11,12)) OR m<-cbind(m,c(10,11,12))
print(m) #
```

```
  [,1] [,2] [,3]
[1,]  1  4  7
[2,]  2  5  8
[3,]  3  6  9
[4,] 10 11 12
```

OR

```
  [,1] [,2] [,3] [,4]
[1,]  1  4  7 10
[2,]  2  5  8 11
[3,]  3  6  9 12
```

iii.Remove row and column. [row,column]

```
m<-matrix(c(1,2,3,4,5,6,7,8,9),nrow=3,ncol=3)
```

```
m<-m[-2,]    OR   m<-m[, -2]
```

```
print(m) #
```

```
  [,1] [,2] [,3]
```

```
[1,]  1  4  7
```

```
[2,]  3  6  9
```

OR

```
  [,1] [,2]
```

```
[1,]  1  7
```

```
[2,]  2  8
```

```
[3,]  3  9
```

iv.Transpose.

```
m<-matrix(c(1,2,3,4,5,6,7,8,9),nrow=3,ncol=3)
```

```
m<-t(m)
```

```
print(m) #
```

```
  [,1] [,2] [,3]
```

```
[1,]  1  2  3
```

```
[2,]  4  5  6
```

```
[3,]  7  8  9
```

v.Special Function.

1.Diagonal matrix.

```
m<-c(1:3)
```

```
m<-diag(m)
```

```
print(m) #
```

```
  [,1] [,2] [,3]
```

```
[1,]  1  0  0
```

```
[2,]  0  2  0
```

```
[3,]  0  0  3
```

2.Upper triangular matrix.

3.Lower triangular matrix.

4.Determinant of matrix.

```
m<-matrix(1:9,nrow=3)
```

```
m<-det(m)
```

```
print(m) #0
```

5.Add/subtract of matrix.

```
x<-matrix(c(1,2,3,4),nrow=2)
```

```
y<-matrix(c(5,6,7,8),nrow=2)
```

```
m<-x+y OR m<-x-y
```

```
print(m) #
```

```
 [,1] [,2]
```

```
[1,]  6  10
```

```
[2,]  8  12
```

OR

```
 [,1] [,2]
```

```
[1,] -4 -4
```

```
[2,] -4 -4
```

6.Multiplication of vectors.

```
x<-matrix(c(1,2,3,4),nrow=2)
```

```
y<-matrix(c(5,6,7,8),nrow=2)
```

```
m<-x%*%y
```

```
print(m) #
```

```
 [,1] [,2]
```

```
[1,] 23 31
```

```
[2,] 34 46
```

7.Inverse of matrix.

```
x<-matrix(c(1,2,3,4),nrow=2)
```

```
print(solve(m)) #
```

```
 [,1] [,2]
```

```
[1,] 11.5 -7.75
```

```
[2,] -8.5  5.75
```

ROLL NO:FYCS56

SUBJECT:DATA SCIENCE (R)

3.Lists.